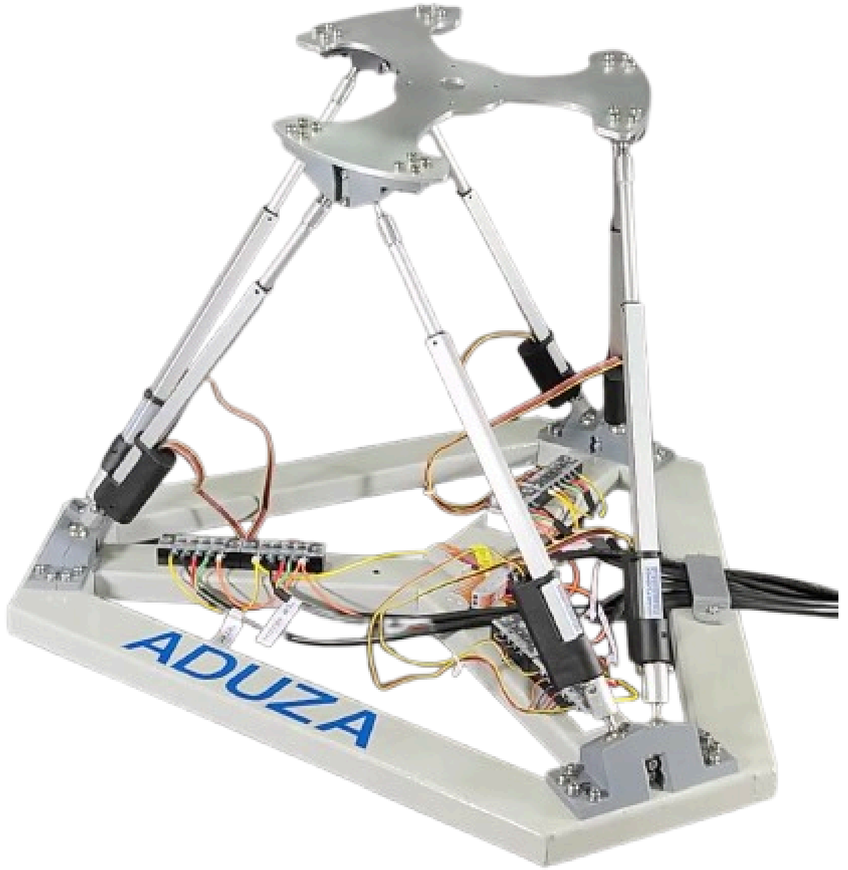
ADUZA Robotics & Automation PROJECT 5

STEWART ULTRASOUND ROBOT



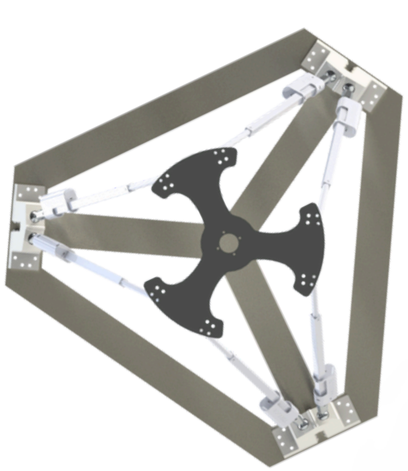
NOTE:

**MY CONTRIBUTION IN THIS PARTICULAR COLLABORATIVE PROJECT WAS ONLY MAKING THE MECHANICAL DESIGN AND ASSEMBLY. I DID NOT BUILD THE CIRCUIT AND CODE HERE.*

STEWART ULTRASOUND ROBOT

BRIEF BACKGROUND

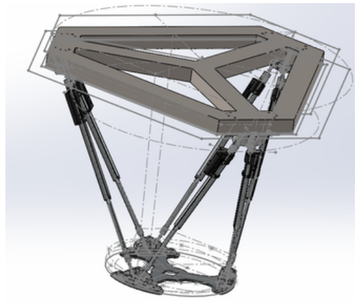
PKMs or parallel robots are the manipulators which having the end effector connected to the ground via several chains having actuations i parallel will have greater rigidity and superior positioning capability. There are various configurations of this platform, but we would be looking at the 6 UPS (or 6 SPS) or commonly called the Stewart platform.
Our goal is to use this Stewart platform for aseptic remote ultrasound procedure. The ultrasound probe will be attached at its end effector and the robot would be controlled via a 6 dof joystick

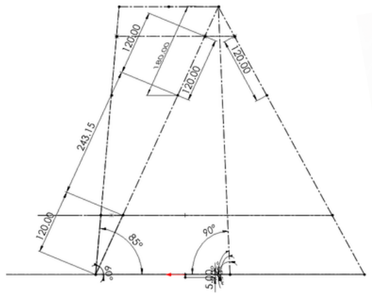




FINALISED CAD MODEL

IDEATION AND FORMULATION

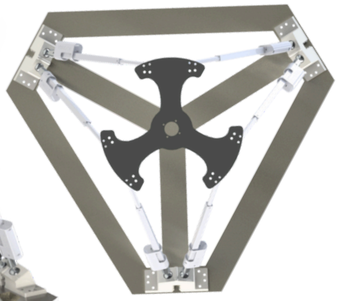


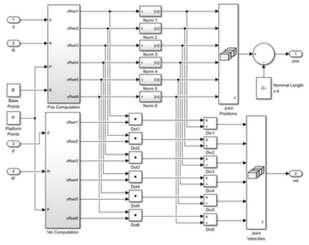


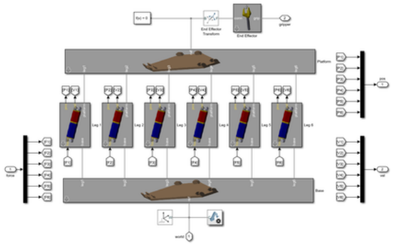
INITIAL CONCEPT

FINAL MODEL

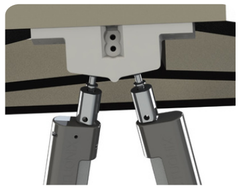




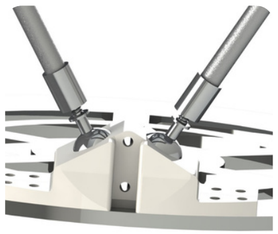


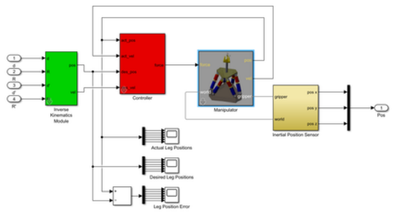


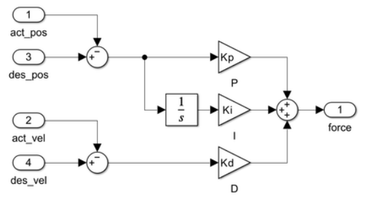
SIMULINK FORMULATION MODEL





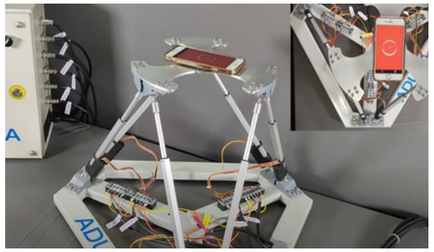


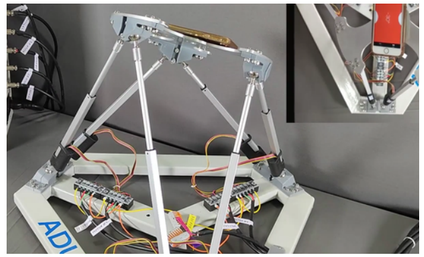


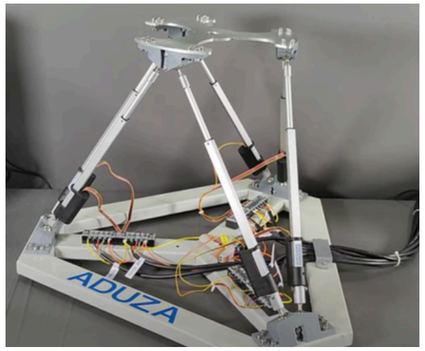


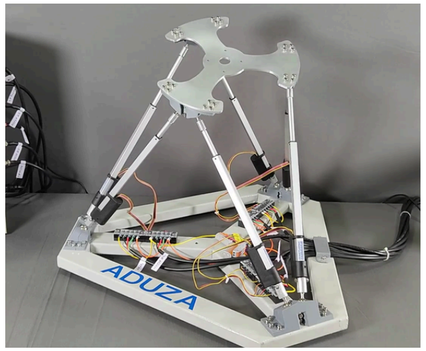
HARDWARE

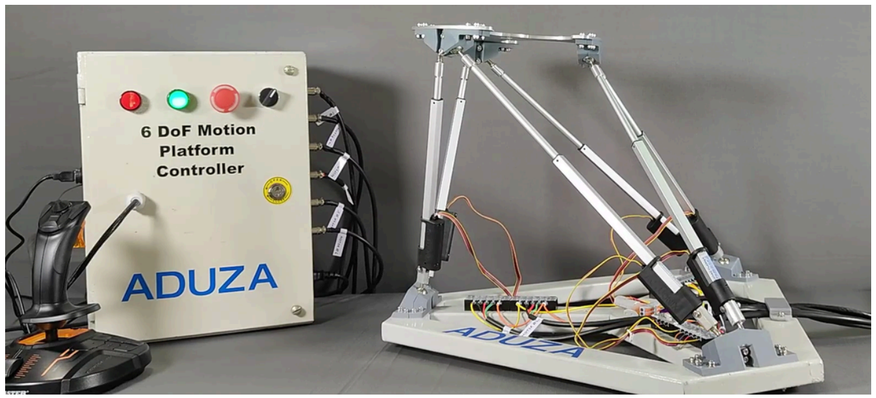
FINAL ASSEMBLY



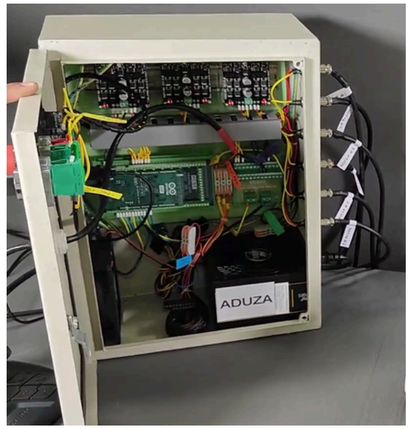


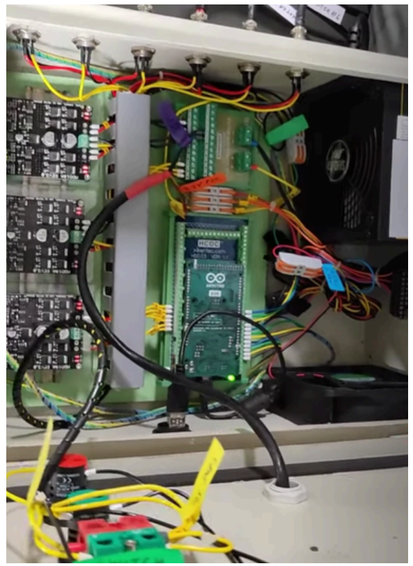






ELECTRONICS





DEPLOYMENT



COMPONENT LIST

- 6 × Linear Actuators
- STM32 Microcontroller Board
- Cooling Fan
- Encoder Feedback
- Industrial Connectors
- 6 × Motor Driver Modules
- Custom Interface PCB
- 24 V DC Switching Power Supply
- Terminal Blocks and Connectors

CONCLUSION

Developed under ADUZA Robotics as a Stewart platform-based ultrasound robot, the final system achieved a workspace of ±150 mm in the x and y directions, 0–220 mm vertical travel, ±30° roll and pitch, and ±60° yaw. The project demonstrated the versatility and dexterity of parallel manipulators while providing valuable experience in kinematic analysis, precision manufacturing, and robotic system integration.